

Statistical Learning Exercise Answers

Chapter 1 and Linear Regression

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August 19, 2026

Reference

These answers are based on *An Introduction to Statistical Learning: With Applications in Python (ISLP)* and the exercises provided for the course. The ISLP book itself labels Linear Regression as Chapter 3, although the course materials may refer to it as Chapter 2.

CHAPTER ONE

Question 1: Flexible versus Inflexible Statistical Learning Methods

(a) Very large sample size and small number of predictors

Answer: A flexible method is generally better.

A very large sample provides enough observations to support a more complex model and reduces the risk of overfitting. Since the number of predictors is small, the model is also relatively easy to estimate.

Memory point: Large n makes flexibility safer.

(b) Very large number of predictors and small number of observations

Answer: An inflexible method is generally better.

When there are many predictors but few observations, a flexible method can fit the training data too closely and overfit. A less flexible method is generally more stable and has lower variance.

Memory point: Small n + large p means avoid excessive flexibility.

(c) Highly non-linear relationship between predictors and response**Answer: A flexible method is generally better.**

A flexible method can capture complicated and non-linear relationships between the predictors and the response, whereas a highly inflexible model may be too restrictive.

Memory point: Non-linear relationship means flexibility helps.

(d) Extremely high error variance**Answer: An inflexible method is generally better.**

High error variance means that the data contain substantial random noise. A highly flexible method may attempt to fit this noise and overfit the training data. A simpler model is generally more stable.

Memory point: High noise means keep the model simple.

Part	Better method	Main reason
(a)	Flexible	Large sample reduces overfitting risk
(b)	Inflexible	Small sample and many predictors increase overfitting
(c)	Flexible	Captures non-linear relationships
(d)	Inflexible	High noise makes flexibility risky

Question 10: Boston Housing Data

(a) Loading the Boston data set

The Python version of ISLP uses the ISLP package. The data can be loaded using:

```
from ISLP import load_data
Boston = load_data("Boston")
```

(b) Number of rows and columns

The Boston data set contains:

506 rows and 13 columns

Each row represents a Boston suburb/census tract. The columns represent characteristics of the suburb, including crime rate, zoning, industrial activity, Charles River location, pollution, number of rooms, housing age, distance to employment centres, highway accessibility, tax rate, pupil-teacher ratio, lower-status population percentage, and median house value.

Variable	Meaning
<code>crim</code>	Per-capita crime rate
<code>zn</code>	Proportion of residential land zoned for large lots
<code>indus</code>	Proportion of non-retail business land
<code>chas</code>	Charles River indicator
<code>nox</code>	Nitrogen oxide concentration
<code>rm</code>	Average rooms per dwelling
<code>age</code>	Proportion of older owner-occupied homes
<code>dis</code>	Distance to employment centres
<code>rad</code>	Accessibility to radial highways
<code>tax</code>	Property-tax rate
<code>prratio</code>	Pupil-teacher ratio
<code>lstat</code>	Lower-status population percentage
<code>medv</code>	Median value of owner-occupied homes

(c) Pairwise scatterplots

The scatterplots show several relationships among the variables. In particular, `rad` and `tax` have a strong positive relationship, while `indus` and `nox` are positively related. `age` and `nox` also tend to be positively related.

There are negative relationships between `dis` and several other variables. In addition, `rm` and `lstat` show a negative relationship. Overall, the scatterplots indicate that several predictors are correlated with one another.

(d) Predictors associated with per-capita crime rate

Yes. Crime rate is associated with several predictors.

The strongest positive associations include `rad`, `tax`, `indus`, `nox`, and `lstat`. Crime is negatively associated with variables such as `dis`, `zn`, and `medv`.

The crime-rate variable ranges from approximately

0.00632 to 88.9762,

and its distribution is strongly right-skewed.

(e) High crime, tax rates, and pupil-teacher ratios

Some suburbs have particularly high crime rates. The range of `crim` is approximately

0.00632 to 88.9762.

The tax rate ranges from

187 to 711.

The pupil-teacher ratio ranges from

12.6 to 22.0.

Crime has much more extreme variation than the pupil-teacher ratio. The crime distribution is strongly right-skewed, with a small number of suburbs having exceptionally high crime rates.

(f) Suburbs bordering the Charles River

The variable `chas` equals 1 when a suburb bounds the Charles River and 0 otherwise.

There are:

35 suburbs

that bound the Charles River.

(g) Median pupil-teacher ratio

The median pupil-teacher ratio is:

19.05

Thus, the median is approximately 19.05 pupils per teacher.

(h) Lowest median value of owner-occupied homes

Two suburbs have the lowest median house value: observations 399 and 406. For both,

$$\text{medv} = 5,$$

which corresponds to \$5,000 because `medv` is measured in thousands of dollars.

These two observations have several unusual characteristics, including very high crime, older housing, high highway accessibility, relatively high taxes, and relatively high pupil-teacher ratios. They also have relatively low numbers of rooms and relatively high values of `lstat`.

Conclusion: The two lowest-value observations have several unfavourable socioeconomic and environmental characteristics.

(i) Suburbs with more than seven and eight rooms

There are:

64

suburbs with more than seven rooms per dwelling.

There are:

13

suburbs with more than eight rooms per dwelling.

The 13 suburbs with more than eight rooms are relatively uncommon and tend to have higher median house values and lower percentages of lower-status population.

LINEAR REGRESSION

Question 1: Hypothesis Tests and p-values

The question concerns the multiple linear regression model relating *Sales* to *TV*, *radio*, and *newspaper* advertising.

For each predictor, the null hypothesis is that the corresponding advertising medium has no association with Sales after accounting for the other advertising media.

TV

H_0 : TV advertising spending has no association with Sales,

after accounting for radio and newspaper advertising.

The p-value is less than 0.0001. Therefore, we reject H_0 .

Conclusion: There is very strong evidence that TV advertising spending is associated with Sales after accounting for radio and newspaper advertising.

Radio

H_0 : radio advertising spending has no association with Sales,

after accounting for TV and newspaper advertising.

The p-value is less than 0.0001. Therefore, we reject H_0 .

Conclusion: There is very strong evidence that radio advertising spending is associated with Sales after accounting for TV and newspaper advertising.

Newspaper

H_0 : newspaper advertising spending has no association with Sales,

after accounting for TV and radio advertising.

The p-value is 0.8599. Since this is greater than 0.05, we fail to reject H_0 .

Conclusion: There is insufficient evidence that newspaper advertising spending is associated with Sales after accounting for TV and radio advertising.

Predictor	p-value	Conclusion
TV	< 0.0001	Significant association
Radio	< 0.0001	Significant association
Newspaper	0.8599	Not significant

Question 6: Least Squares Line

The fitted simple linear regression line is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

The least-squares estimate of the intercept is

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Set $x = \bar{x}$:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}.$$

Substituting the intercept estimate gives

$$\hat{y} = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 \bar{x} = \bar{y}.$$

Therefore, the least-squares regression line always passes through

$$\boxed{(\bar{x}, \bar{y})}.$$

Memory point: At average X , predicted Y equals average Y .

Question 11: Regression Without an Intercept

The exercise generates data according to

$$x \sim N(0, 1), \quad y = 2x + \epsilon,$$

using the specified random seed.

(a) Regress y onto x without an intercept

The model is

$$Y = \beta X + \epsilon.$$

Using the specified data, the results are approximately:

Quantity	Result
$\hat{\beta}$	1.9762
$SE(\hat{\beta})$	0.11695
t -statistic	16.8984
p -value	6.23×10^{-31}

We test

$$H_0 : \beta = 0.$$

Because the p -value is extremely small, we reject H_0 .

Conclusion: There is extremely strong evidence of an association between x and y . The estimated slope is approximately 1.98, close to the true slope of 2.

(b) Regress x onto y without an intercept

The reversed model is

$$X = \beta Y + \epsilon.$$

The results are approximately:

Quantity	Result
$\hat{\beta}$	0.37574
$SE(\hat{\beta})$	0.02224
t -statistic	16.8984
p -value	6.23×10^{-31}

The coefficient changes, but the t -statistic and p -value remain the same.

(c) Relationship between (a) and (b)

	Y on X	X on Y
Coefficient	1.9762	0.37574
Standard error	0.11695	0.02224
t -statistic	16.8984	16.8984
p -value	6.23×10^{-31}	6.23×10^{-31}

Thus,

$$t_{Y|X} = t_{X|Y}$$

and

$$p_{Y|X} = p_{X|Y}.$$

The direction of regression changes the coefficient because the variables have different scales, but it does not change the strength of evidence against the null hypothesis.

(d) Derivation of the t -statistic

For regression of Y onto X without an intercept,

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}.$$

The standard error is

$$SE(\hat{\beta}) = \sqrt{\frac{\sum_{i=1}^n (y_i - x_i \hat{\beta})^2}{(n-1) \sum_{i=1}^n x_i^2}}.$$

Let

$$A = \sum x_i y_i, \quad B = \sum x_i^2.$$

Then

$$\hat{\beta} = \frac{A}{B}.$$

Expanding the residual sum of squares gives

$$\sum (y_i - x_i \hat{\beta})^2 = \sum y_i^2 - \frac{A^2}{B}.$$

Therefore,

$$SE(\hat{\beta}) = \frac{\sqrt{B \sum y_i^2 - A^2}}{\sqrt{n-1} B}.$$

Hence,

$$t = \frac{\sqrt{n-1} \sum_{i=1}^n x_i y_i}{\sqrt{(\sum_{i=1}^n x_i^2) (\sum_{i=1}^n y_i^2) - (\sum_{i=1}^n x_i y_i)^2}}.$$

(e) Equality of the t-statistics

The formula above is symmetric in x and y . Since

$$\sum x_i y_i = \sum y_i x_i,$$

switching x and y leaves both the numerator and denominator unchanged.

Therefore,

$$\boxed{t_{Y \text{ on } X} = t_{X \text{ on } Y}}.$$

(f) Regression with an intercept

When an intercept is included, the t-statistic for

$$H_0 : \beta_1 = 0$$

is also the same whether Y is regressed on X or X is regressed on Y .

In simple linear regression with an intercept,

$$t = r \sqrt{\frac{n-2}{1-r^2}},$$

where r is the sample correlation.

Because

$$\text{Cor}(X, Y) = \text{Cor}(Y, X),$$

the t-statistic is identical in both regression directions.

$$\boxed{t_{Y \text{ on } X} = t_{X \text{ on } Y}}$$

Final Revision Points

- Small p-value $\Rightarrow$ reject H_0 .
- Large p-value $\Rightarrow$ fail to reject H_0 .
- In simple linear regression, the least-squares line passes through $(\bar{x}, \bar{y})$.
- Flexible methods are useful when there is enough data or a complicated non-linear relationship.
- Inflexible methods are safer when the sample is small or the data contain substantial noise.

- Reversing simple regression changes the coefficient, but the t-statistic for a zero-slope test remains the same.